

AN AUTOFOCUS METHOD FOR A TEM

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Autofocus and correction of astigmatism of a transmission electron microscope (TEM) based on measuring a beam-tilt-induced image displacement is proposed and its theoretical limitations are studied. Tilting the illumination beam displaces the specimen image on screen when the TEM is out of focus. This displacement has a known relationship with the defocus. Autofocusing is possible by tilting the beam, measuring the image displacement, calculating the defocus and correcting it. Correction of astigmatism is possible by measuring the defocus in different directions. The method is fast because it calculates and corrects the defocus in one step. It works with many types of specimens because it utilizes both the amplitude and phase contrast of a bright field image. The precision of this method depends on the precision of the image displacement estimation. The shifted and unshifted images differ because of shot noise, instrumentation noise, and aberrations caused by the beam tilt. An expression is derived, containing parameters of the TEM and measuring system, for the achievable precision in estimating the displacement. This expression is a tool for optimizing the automatic focussing procedure and the measuring system. It does not depend on any particular estimation method with which the displacement is calculated. Computer simulations for a TEM equipped with a Vidicon videocamera have been carried out. They show that at Scherzer defocus (86 nm) the minimum measuring time required for focussing the TEM with a precision of 5 nm is about 50 ms. The precision is less satisfactory (> 30 nm) when, with the same measuring time, the TEM is far out of focus or very near focus. The precision improves proportionally to the square root of the measuring time.

1. Introduction

Reliably interpretable high-resolution electron micrographs require an accurately focussed, stigmated and aligned transmission electron microscope (TEM). Incorrect objective lens and stigmator currents result in artifacts in the recorded image. Focussing and stigmating a TEM can be time consuming and requires an experienced operator, in the case of low-dose and high-resolution work in particular. The time factor is important since object structures are destroyed by prolonged radiation. A system which automatically focusses, stigmates and aligns a TEM is useful when it is:

- (1) as *precise* as or more precise than a human operator;
- (2) as *fast* as or faster than a human operator;
- (3) *efficient*, in the sense that it uses the available observations exhaustively;
- (4) *reproducible* (as is not the case with human operators);

- (5) *applicable with most specimens*;
- (6) *suitable to manufacture and to implement* in a TEM at a reasonable cost.

Several methods to automatically focus, stigmatize and align a TEM have been proposed [1]. To our knowledge, the only published proposal for automatic focussing which has been tested in practice is based on minimum contrast of the image at focus [2]. Image contrast is measured as a function of the objective lens current. The object is in focus when the image has the lowest contrast. This minimum contrast is used by most microscopists while focussing, as it involves no special equipment. The same method is also applied for alignment of the illuminating beam [3]. Another method is based on the power spectrum of the image (diffractogram) [4]. The position of peaks and zeros in the spectrum make it possible to compute the defocus, astigmatism and misalignment. Unfortunately, there is considerable ambiguity in the imaging conditions which might be

deduced from a single diffractogram. This limits the practical feasibility of the method.

2. Focussing by measuring beam-tilt-induced image displacement

The method for fast and direct defocus calculation considered in this paper is based on tilting the illuminating electron beam over a known angle, thus producing image displacement on the screen because of the large depth of field due to the very coherent illumination. This beam-tilt-induced image displacement (BID) has a known relationship with the defocus of the TEM [5] and can be used as focussing aid, see fig. 1. Le Poole [6] introduced in 1947 the beam tilt wobbler as a simple focussing aid. Most microscopists use this method for focussing a TEM at lower magnifications. Later LePoole and De Groot [7] proposed a special linear detector to measure the direction of the image displacement. A similar approach was followed by Curling et al. [8] using only one detector. Koops and coworkers [9,10] measured the aberrations of the objective lens with an aplanator. He used a special specimen for his measurements: a specimen with one very small hole, whose position was detected by scanning its image over a single detector.

Van der Mast [11] proposed to measure both magnitude and direction of the image displacement in order to calculate and correct the defocus in one step: the BID method. It uses a multi-element image pick-up system and an algorithm which calculates the image displacement directly, independently of the type of specimen. In the case of a properly aligned system the image displacement D for beam tilt angles $+\beta$ and $-\beta$ (as seen from the specimen) is given by McFarlane [5], and Zemlin et al. [12]

$$D = 2M\beta \left(\Delta f + C_s \beta^2 + C_s \omega^2 \frac{2\lambda^2}{4\pi^2} \right), \quad (1)$$

with M the magnification, Δf the defocus of the TEM, C_s the spherical aberration of the objective lens, λ the wavelength of an accelerated electron and ω the spatial frequency of the specimen ex-

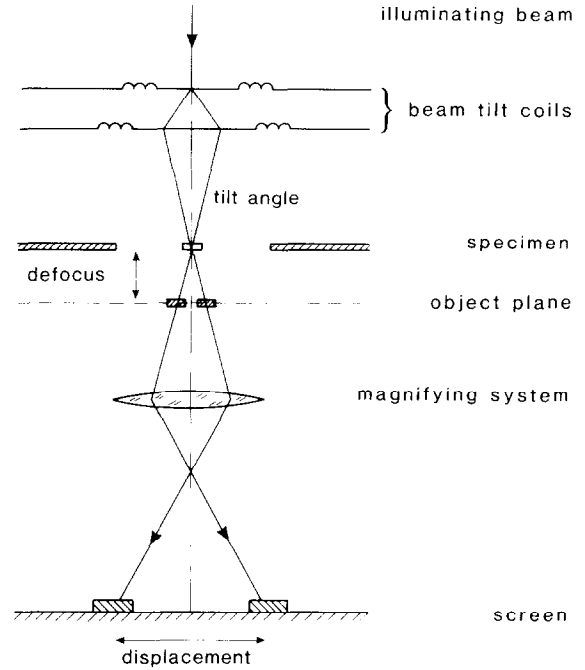


Fig. 1. Image displacement caused by beam tilt.

pressed in rad m^{-1} . For low frequencies and for large tilt angles the frequency-dependent term is negligible. The displacement is then given by:

$$D = 2M\beta(\Delta f + C_s \beta^2). \quad (2)$$

The BID method is suitable as an autofocussing system when the requirements mentioned in the introduction are met. We will first discuss its precision.

3. Achievable precision in estimating an image displacement

3.1. Model of the observations

The specimen is supposed to be imaged with axial, partly coherent illumination and producing phase contrast images. These assumptions are valid for high-resolution electron microscopic images when the beam tilt angle is not too large. Observations are made at the image of the specimen on the fluorescent screen at every point of a given interval of an x -axis. Measurements are done during T

seconds using a rectangular sensor with length Δx and width Δy (scaled to the specimen plane). The width Δy is supposed to be small enough to presume the image measured in the y -direction constant. This is done for simplicity; the theory can be extended to correct for this assumption. Measuring the image with the beam tilted first over an angle β and next over $-\beta$ results in the following observations:

$$r_1(x) = k(x) + n_1(x), \quad (3a)$$

$$r_2(x) = k(x - D) + n_2(x), \quad (3b)$$

with x the position of a sensor along the x -axis, $k(x)$ the image of the specimen, $k(x - D)$ the displaced image over a distance D , and $n_1(x)$ and $n_2(x)$ independent, zero-mean, Poisson-distributed noises with expectation

$$k = \Delta x \Delta y \bar{T} \bar{k}, \quad (4)$$

where k is the total number of electrons passing through an area equal to $\Delta x \Delta y$ with $\bar{k}$ the number of electrons per second and per square meter (averaged over all sensors). The expectation of the Poisson noise can be considered independent of the coordinate x because of the low contrast in electron microscopic images (0–0.2) at high magnifications. Furthermore, the noise is considered normally distributed as the number of electrons counted per sensor is large with respect to its square root.

This model of the observations (3) is next used to derive an expression for the achievable precision with which the displacement can be estimated irrespective of the estimator used. In statistics the expression derived is called minimum variance bound (MVB) or Cramér–Rao lower bound (Van den Bos [13]). The MVB is used for two purposes: (1) from it numerical values can be computed for the achievable precision of the measured image displacement; (2) the expression for the MVB shows the dependence of this precision on experimental conditions as defocus, image recording strategy, instrumentation noise and measuring time. This dependence is investigated numerically for various realistic experimental conditions in section 4.

3.2. Minimum variance bound

The minimum variance bound (MVB) is a lower bound on the variance of any unbiased estimator. In this particular case, the MVB of a one-dimensional image displacement has to be derived. This one-dimensional approach makes it possible to treat the problem as a time-delay-estimation problem in passive sonar theory, for which the MVB already has been derived by Carter [14] and is described by the following expression:

$$\sigma_D^2 \geq 2\pi \left[L \int_{-\omega_h}^{\omega_h} \frac{\eta^2(\omega)}{1 + 2\eta(\omega)} \omega^2 d\omega \right]^{-1}, \quad (5)$$

where $\eta(\omega)$ is the signal-to-noise ratio (SNR) defined by

$$\eta(\omega) = S_m(\omega) / S_n(\omega), \quad (6)$$

with ω_h the highest frequency present in the image $k(x)$, L the length of the interval on which observations are made, $S_m(\omega)$ the power spectrum of $k(x)$ and $S_n(\omega)$ the power spectrum of both the noises $n_1(x)$ and $n_2(x)$. The expressions (5) and (6) show the importance of the SNR $\eta(\omega)$ at high frequencies. When the SNR is large for high frequencies the MVB will be small. When the SNR is small for high frequencies the MVB will be large. Characteristics of the TEM and measuring set-up influence $\eta(\omega)$ strongly and are studied in more detail below.

3.3. Signal-to-noise ratio, Poisson noise

In this section the SNR of the measured image is derived as a function of parameters describing the image formation on screen, and parameters describing the measuring system. The image formation at high magnification can be described with the phase contrast transfer function (PCTF). Several PCTFs are described in the literature [15]. A well known PCTF given by Reimer (see Hanszen [16]) includes the effect of partly coherent axial illumination of the specimen and of the energy spread of the electrons:

$$\begin{aligned} H(\omega; a) = & -2 \sin[a_1 \omega^4 - a_2 \Delta f \omega^2] \\ & \times \exp\left[-(a_3 \omega^2)^2\right] \\ & \times \exp\left[-(a_4 \omega^3 - a_5 \Delta f \omega)^2 \theta^2\right], \quad (7) \end{aligned}$$

with

$$a_1 = \frac{C_s \lambda^3}{32\pi^2}, \quad a_2 = \frac{\lambda}{4\pi}, \quad a_3 = \frac{\lambda H_A}{16\pi(\ln 2)^{1/2}},$$

$$a_4 = \frac{C_s \lambda^2}{8\pi^2(\ln 2)^{1/2}}, \quad a_5 = \frac{1}{\pi(\ln 2)^{1/2}},$$

with θ the angle of the beam seen from the specimen, $a = (a_1 \dots a_5)$ the vector of specifications of the TEM.

$$H_A = C_c \frac{\Delta V}{V} \left(\frac{1 + E/E_0}{1 + E/2E_0} \right), \quad (7a)$$

with C_c the chromatic aberration, $\Delta V/V$ the stability of the acceleration voltage, E_0 the rest-energy of an electron and E the energy of an accelerated electron. The envelope function of $H(\omega; a)$ related to the chromatic aberration is dominating and is reduced to 1% or less of its maximum value for frequencies higher than:

$$\omega_h = (2/\lambda H_A)^{1/2}, \quad (8)$$

where H_A is described by (7a). Fig. 2 shows the PCTF for the specifications in table 1 and for a defocus of $\Delta f = 2$ Sch.

The relationship between the power spectrum of the image on the screen $S_i(\omega)$ and the power spectrum of the specimen $S_s(\omega)$ is now given by

$$S_i(\omega) = |H(\omega; a)|^2 S_s(\omega). \quad (9)$$

The relationship of the image on the screen $S_i(\omega)$ with the power spectrum of the image measured is (see appendix A):

$$S_m(\omega) = T^2 (\Delta x)^2 (\Delta y)^2 \text{sinc}^2(\omega \Delta x/2) S_i(\omega), \quad (10)$$

where

$$\text{sinc}(p) = \frac{\sin(p)}{p}. \quad (11)$$

In the following it is convenient to normalize $S_s(\omega)$ with respect to the power spectrum on screen $S_m(\omega)$. This is done as follows.

Define the modulation depth as:

$$m = \sigma_k/k, \quad (12)$$

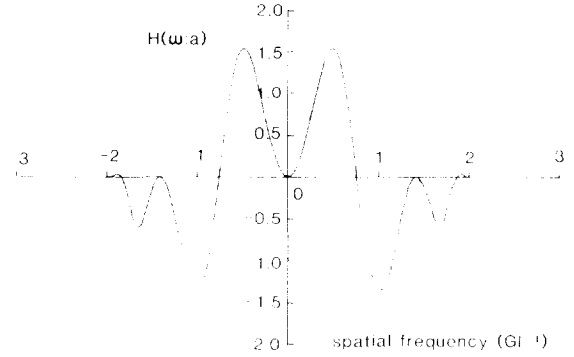


Fig. 2. The phase contrast transfer function. See table 1 for specifications of the TEM. $\Delta f = 2$ Sch.

where σ_k is the standard deviation of number of electrons counted per sensor (standard deviation of the image). Now suppose that at Scherzer defocus ($\Delta f = (C_s \lambda)^{1/2}$), and for a particular specified sensor length Δx_s , m has the value μ . Generally, σ_k must satisfy

$$\sigma_k^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_m(\omega) d\omega. \quad (13)$$

Hence, from (3), (9) and (10)

$$m^2 = \frac{1}{2\pi k^2} \int_{-\infty}^{\infty} \text{sinc}^2(\omega \Delta x/2) |H(\omega; a)|^2 \times S_s(\omega) d\omega. \quad (14)$$

Hence, in order to produce a modulation depth $m = \mu$ at Scherzer defocus with $\Delta x = \Delta x_s$, $S_s(\omega)$ must satisfy

$$S_s(\omega) = \mu^2 k^2 S_{ns}(\omega), \quad (15)$$

Table 1

Microscope parameters and measuring set-up; dimensions are those at specimen level; 1 Sch = 86 nm, 1 Gl = 0.56 nm

Specimen	TEM	Measuring set-up
$\mu = 0.05$	$C_s = 2$ mm	$\Delta x = 0.05$ nm
	$C_c = 2$ mm	$\Delta y = 0.5$ nm
Flat power spectrum	$V = 100$ kV	$L = 0.25$ μ m
	$M = 10^5$	
	$k = 10^{21}$ electrons $\text{m}^{-2} \text{s}^{-1}$	
	$\beta = 5$ mrad	
	$\theta = 1$ mrad	

with

$$\frac{1}{2\pi} \int_{-\omega_h}^{\omega_h} \text{sinc}^2(\omega \Delta x_s/2) |H(\omega; a)|_{\text{Scherzer}}^2 \times S_{ns}(\omega) d\omega = 1, \quad (16)$$

and $S_{ns}(\omega)$ the normalized power spectrum of the specimen.

Of all spectra having the *shape* as $S_s(\omega)$, only one normalized spectrum $S_{ns}(\omega)$ gives rise to the supposed modulation depth μ for the given sensor length Δx , number of electrons counted per sensor k and PCTF $H(\omega; a)$. The power spectrum of the image $k(x)$ is therefore determined by the modulation depth μ of the specimen, the shape of the power spectrum of the specimen $S_s(\omega)$, the PCTF, the illumination $\bar{k}$, the measuring time T and sensor dimensions Δx and Δy .

It is shown in appendix B that the power spectrum of the Poisson noise $n(x)$ is described by:

$$S_{pn}(\omega) = (\bar{k}T) (\Delta x)^2 \Delta y \text{sinc}^2(\omega \Delta x/2). \quad (17)$$

Clearly the Poisson noise depends on the illumination, measuring time and the dimensions of the sensor.

It follows from (6), (10) and (17) that the SNR $\eta(\omega)$ is described by

$$\eta(\omega) = \mu^2 (\bar{k}T) \Delta y |H(\omega; a)|^2 S_{ns}(\omega). \quad (18)$$

This expression shows that the SNR $\eta(\omega)$ depends on the specimen ($S_s(\omega)$ and μ), the PCTF and measuring strategy ($\bar{k}$, T and Δy). It also shows that, under the assumption that the bandwidth of the specimen is larger than that of the PCTF, the frequency ω_h in (5) may be taken as the frequency at which the PCTF becomes negligible and remains so. Notice that the sensor length Δx is absent in expression (18). It could be concluded that the sensor length is of no importance for the achievable precision! In the next section it will be shown that this is no longer true when instrumentation noise is present in the model of the measured image and a finite number of sensors is used.

3.4. Instrumentation noise

Up to now it has been assumed that there is no instrumentation noise present in the image measured. Of course, in practice, there is always instrumentation noise present as, for instance, digital quantization noise or dark current in a video camera. Suppose there is additive instrumentation noise $n_i(x)$, not correlated with the Poisson noise $n_p(x)$. The total noise in the image can then be modelled by

$$S_n(\omega) = S_{pn}(\omega) + S_{in}(\omega), \quad (19)$$

with $S_{in}(\omega)$ the power spectrum of the instrumentation noise $n_i(x)$. To be able to handle the instrumentation noise in the same way as the Poisson noise, it is assumed that the instrumentation noise is normally distributed and that its power spectrum may be modelled by

$$S_{in}(\omega) = S_{ni} \text{sinc}^2(\omega \Delta s/2), \quad (20)$$

where the constants S_{ni} and Δs define the intensity and the bandwidth of $S_{in}(\omega)$. To model (approximately) white instrumentation noise before and after sampling, Δs is small compared to Δx , so that $S_{in}(\omega) \approx S_{ni}$ over the bandwidth of PCTF, and band-limited. This finite power spectrum is of importance when (in the next subsection) sampling is introduced. Using (6) and (19), the SNR $\eta_i(\omega)$ is now given by:

$$\eta_i(\omega) = N(\omega) \eta_p(\omega), \quad (21)$$

where

$$N(\omega) = S_{pn}(\omega) / [S_{in}(\omega) + S_{pn}(\omega)], \quad (22)$$

$$\eta_p(\omega) = S_m(\omega) / S_{pn}(\omega). \quad (23)$$

Consider the factor $N(\omega)$ in (21); the envelope of $N(\omega)$ is less than 1 and the sensor length affects the function $N(\omega)$ (and so the SNR $\eta_i(\omega)$), but not its envelope, as can be seen in fig. 3. The number of electrons per square meter $\bar{k}T$ relative to S_{ni} and the sensor dimensions Δx and Δy are now also of importance for the MVB, see (17), (20) and (22). The influence of instrumentation noise on the MVB is minimal when the zeros of

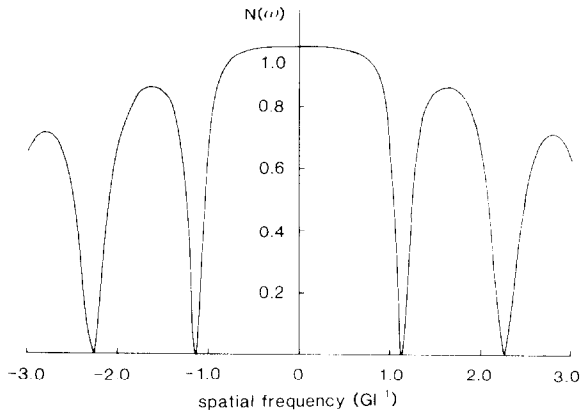


Fig. 3. Influence of instrumentation noise $N(\omega)$ on the SNR. See tables 1 and 2 for specifications of the TEM and measuring set-up. $1 \text{ Gl} = 0.56 \text{ nm}$.

the PCTF coincide as much as possible with the zeros of the function $N(\omega)$.

3.5. Sampling and its influence on the MVB

The image measurements are supposed to be made using a finite number of sensors. This is, in fact, sampling of the image, and it influences the power spectra $S_m(\omega)$ and $S_n(\omega)$. To what extent this is the case is determined by the sampling frequency ω_s , the bandwidth of the spectra, the sensor length and the spectrum of the instrumentation noise. An expression for the relationship of the power spectrum of a process before and after sampling is given in ref. [17]. In terms of $S_m(\omega)$ this relationship yields:

$$S'_s(\omega) = \sum_{n=-\infty}^{\infty} S_m(\omega + n\omega_s), \quad (24)$$

with

$$\omega_s = 2\pi/\Delta\xi, \quad (25)$$

where $S'_s(\omega)$ is the spectrum of $k(x)$ after sampling and $\Delta\xi$ is the sensor distance. The expression for the power spectrum of the sampled version of the Poisson noise is analogous to (24). Substituting (17) into this expression and subsequently applying Poisson's sum formula [17], one

can easily show that

$$S'_{pn}(\omega) = \bar{k}T\Delta x\Delta y\Delta\xi \left[1 + 2 \sum_{n=1}^{\infty} d_1(n\Delta\xi) \times \cos(n\omega\Delta\xi) \right], \quad (26a)$$

$$d_1(x) = \begin{cases} 1 - |x|/\Delta x & \text{for } |x| \leq \Delta x, \\ 0 & \text{elsewhere.} \end{cases} \quad (26b)$$

When the sampling distance equals the sensor length the sampled power spectrum of the Poisson noise after sampling is flat and given by

$$S'_{pn}(\omega) = k\Delta\xi. \quad (27)$$

Similarly, the power spectrum of the sampled instrumentation noise

$$S'_{in}(\omega) = S_m \frac{\Delta\xi}{\Delta s} \left[1 + 2 \sum_{n=1}^{\infty} d_2(n\Delta\xi) \cos(n\omega\Delta\xi) \right] \quad (28a)$$

where

$$d_2(x) = \begin{cases} 1 - |x|/\Delta s & \text{for } |x| \leq \Delta s, \\ 0 & \text{elsewhere.} \end{cases} \quad (28b)$$

The SNR after sampling is now

$$\eta'_i(\omega) = N'(\omega)\eta'_p(\omega), \quad (29)$$

with

$$\eta'_p(\omega) = S'_s(\omega)/S'_{pn}(\omega), \quad (30)$$

$$N'(\omega) = S'_{pn}(\omega)/[S'_{in}(\omega) + S'_{in}(\omega)]. \quad (31)$$

Using formula (5) and substituting $\eta'_i(\omega)$ for $\eta(\omega)$, the influence of the various parameters on the MVB can be studied. A number of such numerical experiments are described in the next section.

4. Numerical experiments

The influence of relevant parameters in the measuring process is not directly clear from (5), (24), (26) and (28). Therefore, numerical experiments have been carried out to study the dependence of the MVB on some of these parameters (defocus, sampling distance, sensor length, instru-

mentation noise, magnification and measuring time, see table 2) in order to ascertain their influence on the precision with which the defocus can be estimated.

Table 1 shows the microscope characteristics and measurement parameters used (the reference conditions). It is supposed that the specimen has a flat power spectrum (up to ω_h or ω_s) and a modulation depth μ of 0.05. The specifications of the TEM are those of a Philips EM 420: $C_c = 2$ mm, $C_s = 2$ mm, stability of the acceleration voltage $\Delta V/V = 10$ ppm. The acceleration voltage is 100 kV. The screen current density is 16 nA m^{-2} (10^{21} electrons $\text{m}^{-2} \text{ s}^{-1}$). The angle of the beam seen from the specimen is 1 mrad. The defocus is expressed in Sch [18] and in m. For these specifications of the TEM 1 Sch is about 86 nm. The tilt angle is 5 mrad, small enough to allow the use of the expression in Reimer [15] for the PCTF of an axially illuminated specimen. The magnification influences the scaling from screen to specimen level; chosen is 10^5 . The measuring set-up simulated is one line of a videocamera with a noise intensity half the Poisson noise intensity for the reference conditions. A pixel of the video image is considered as sensor with dimensions $\Delta x \Delta y$. The measuring time is 5 s. The length of the measuring interval is 2.5 cm and the width and length of a pixel on screen of $50 \mu\text{m}$.

To make the results easier to interpret, not the MVB for estimation of the displacement is plotted but three times the square root of the corresponding MVB for estimation of the defocus:

$$3\sigma_{\Delta f} = 3(\text{MVB})^{1/2}/(2\beta). \quad (32)$$

This can be interpreted as the 99.5% confidence level for estimation of the defocus. In what follows (32) will be referred to as precision.

4.1. Defocus

The precision of focussing depends on the image modulation depth (or contrast). Since the latter is a function of the defocus (fig. 4), the precision is a function of the defocus. Fig. 5, calculated from (5) and (32), shows this. Notice the global minimum near Scherzer defocus and the local maximum

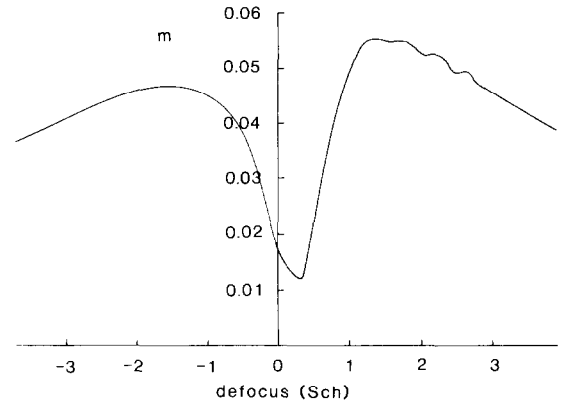


Fig. 4. Modulation depth m of the image on screen as a function of defocus Δf . 1 Sch = 86 nm.

near focus. The figure also shows that if the absolute defocus is less than 1 Sch, or more than 5 Sch, estimates of the defocus may be very imprecise. Then autofocussing the TEM on the basis of image displacement estimates no longer makes sense.

The smallest focus step in a Philips EM 420 is 5 nm. To obtain this accuracy using the BID method a measuring time of more than 5 s is necessary, using only one line of the measured image for calculating the image displacement. When more independent lines are used in calculating the image displacement, the required measuring time decreases proportionally with the square root of the number of lines. The strategy in focussing the

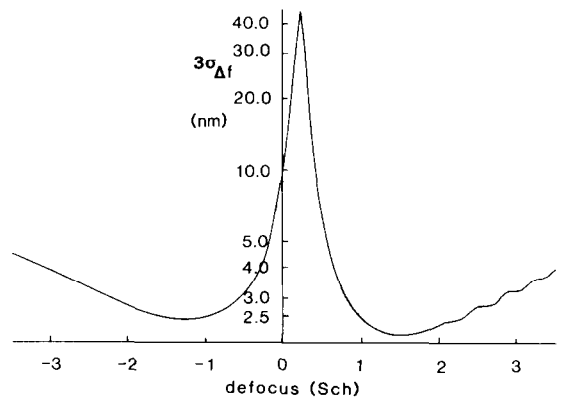


Fig. 5. Precision of the estimation of defocus as a function of defocus. 1 Sch = 86 nm.

TEM can efficiently make use of the relationship between the precision in estimating the defocus and the defocus itself by focussing the TEM in two steps. During the first step the defocus of the TEM is estimated, and the TEM is set at about Scherzer defocus. During the second step the defocus is again estimated and this time the TEM is focussed more precisely.

4.2. Sampling distance

Results related to the sampling distance are presented in fig. 6. The length of the individual sensor elements is set to 0.5 nm. The sampling distance is varied from about one fifth of the sensor length to 5 times the sensor length. When the sampling distance is less than 0.2 nm hardly any new information is gathered, while for sampling distances of more than 1 nm only the phase contrast at low frequencies is used. So a relatively smooth changing of the MVB for both large and small sampling distances is to be expected. A sampling distance of 0.5 nm seems to be optimum under the reference conditions. The smallest sampling distance in a videocamera equals the sensor length, in this case 0.5 nm.

4.3. Sensor length

The sensor length is varied from 0.1 nm up to 3 nm, see fig. 7. This figure shows that the precision in estimating the defocus varies periodically with the sensor length. For very small sensor lengths the SNR is small because of the few electrons

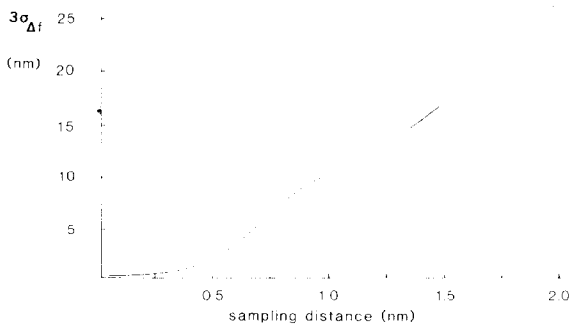


Fig. 6. Precision of the estimation of defocus as a function of sampling distance $\Delta\xi$. Sensor length is 0.5 nm.

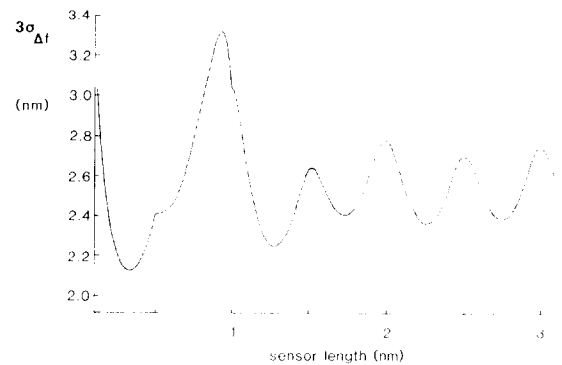


Fig. 7. Precision of the estimation of defocus as a function of sensor length.

counted per sensor. The estimation of the image displacement is therefore imprecise. The peaks in the figure can be understood from (18), (21) and (22): The SNR with respect to the Poisson noise has the same shape as the PCTF (see fig. 8) and the sensor length determines which frequencies of the SNR $\eta(\omega)$ are suppressed by the transfer function $N(\omega)$. The dips in $N(\omega)$ are periodic with Δx (fig. 3), so the MVB will increase (while the SNR decreases) periodically with Δx . By adjusting the sensor length, optimal measuring conditions can be created.

4.4. Instrumentation noise

The influence of the instrumentation noise on the precision is described by (21), (22) and (23)

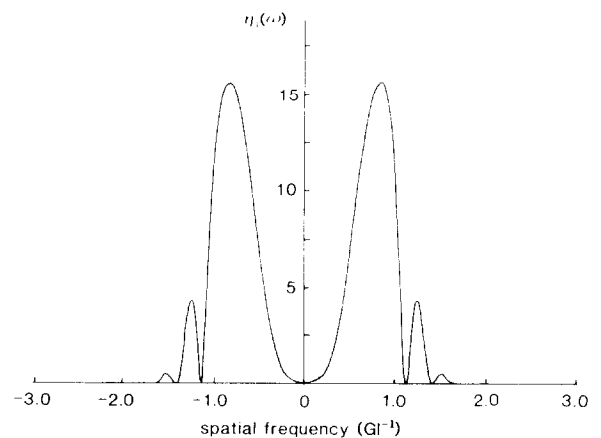


Fig. 8. Signal-to-noise ratio $\eta_1(\omega)$ before sampling. 1 Gl = 0.56 nm.

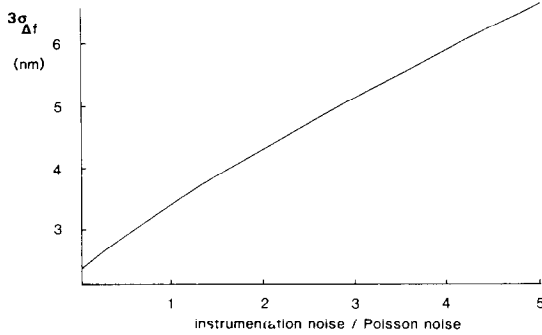


Fig. 9. Precision of the estimation of defocus as a function of the ratio of the Poisson noise to the instrumentation noise.

combined with (5). The ratio of intensities of the instrumentation noise S_{ni} to that of the Poisson noise is varied from 0.1 up 5, and the precision varies as shown in fig. 9. This result can be explained as follows. From (22) it follows that $N(\omega)$ and, therefore, the SNR $\eta_i(\omega)$ described by (21) decreases monotonously with S_{ni} . Since the integrand of (5) decreases monotonously with the SNR, the precision increases monotonously with S_{ni} . Furthermore, if the $\text{SNR} \ll 1$ the integrand is approximately quadratic in the SNR. For the $\text{SNR} \gg 1$ it is linear. This is also in agreement with fig. 9. So, not surprisingly, the instrumentation noise should be kept as small as possible.

4.5. Magnification

The influence of the magnification on the precision can be seen in fig. 10. The magnification influences the dimensions of a sensor, the sam-

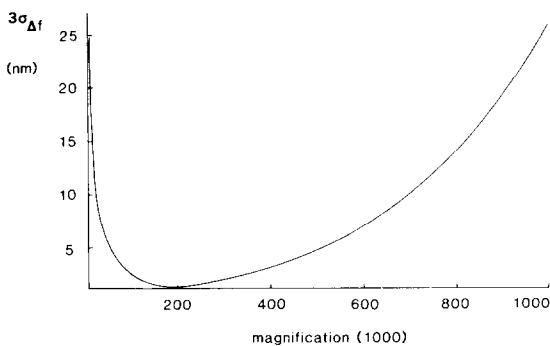


Fig. 10. Precision of the estimation of defocus as function of magnification.

pling distance and the interval of the image in which measurements can be done on specimen level. The total effect of these four parameters on the precision while varying the magnification is rather complex, see (24), (26), (29) and (5). For low magnifications the estimation of the image displacement is relatively imprecise because of the large dimensions of Δx , Δy and $\Delta \xi$ on specimen level. For high magnifications the precision is also less because of the very small dimensions Δx , Δy and $\Delta \xi$, which results in a small SNR, see section 4.3. The magnification best suited for estimating an image displacement is about 200,000 times.

4.6. Measuring time

The measuring time influences the signal-to-Poisson-noise ratio $\eta_p(\omega)$ following (18). The precision is plotted against the measuring time in fig. 11. The precision increases inversely with the square root of the measuring time. The influence of the measuring time on the precision is described by (18), (21) and (22) combined with (5). Expression (18) shows that $\eta_p(\omega)$ is proportional to T . Furthermore, it follows from (17) and (22) that the function $N(\omega)$ monotonously increases to one if T increases. So, by (21), $\eta_i(\omega)$ is somewhat more than proportional to T . Then, by the same arguments as used in section 4.4, the MVB, eq. (5), is more than inversely proportional to T for small T , and becomes for increasing T purely inversely proportional. This is precisely the type of behaviour shown by fig. 11.

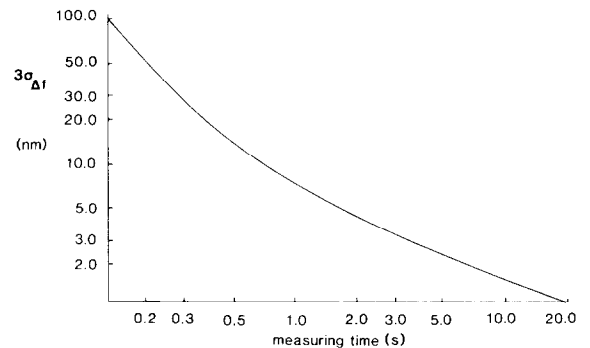


Fig. 11. Precision of the estimation of defocus as a function of measuring time T .

The minimum time necessary to autofocus the TEM up to a certain precision can be calculated using the MVB. In the following it is assumed that precision better than 5 nm is desired. Fig. 6 shows that a measuring time of 5 s is enough, using one line of a video camera, provided the TEM is not more than 3.5 Sch out of focus. When more (independent) video lines are used for calculating the image displacement, the required measuring time is less. Moreover, the precision is inversely proportional to the square root of the number of statistically independent lines used:

$$3\sigma_D \sim 1/n^{1/2}. \quad (33)$$

When 500 independent lines are available ($n = 500$), the minimum measuring time will be about 25 ms instead of 5 s when one line is used to calculate the image displacement. In order to autofocus using the BID method, two measurements have to be done. So the minimum time required is 50 ms. To measure other aberrations (such as astigmatism and misalignment) more images have to be measured [10]. Of course, the time necessary to measure these aberrations increases linearly with the number of image displacements to be estimated.

5. The BID method as autofocus method

We now return to the feasibility requirements to be met by BID autofocus listed in section 1:

(1) As to the *precision*: we have assessed in sections 4 and 5 the ultimate precision than can be achieved with a particular set of observations, automatically or manually. So, any automatic method getting near this precision is unlikely to be outperformed by a human operator.

(2) The BID method is inherently *fast*, because it calculates the defocus directly from two measured images; there are, in principle, no intermediate iterative steps. Limitations in speed depend on the measuring system used.

(3) A practical and natural measure for the *efficiency*, used in statistical literature, is the extent to which an estimation method achieves the minimum variance bound. Methods are available

for the construction of estimation methods optimal in this sense [13]. These methods can, in principle, be applied to the problem of image displacement [19]. However, this has not yet been tried out in TEM practice. Several straightforward, non-optimal methods such as cross-correlation, minimum square error, and determination of the phase angle of the cross-power spectrum between the images are available [20]. Whichever of these methods is chosen, all are based on straightforward signal processing operations as correlation or discrete Fourier transform. An example of an estimator, which has been used in the first experiments by Van der Mast [11], is relatively fast and easy to implement in hardware:

$$F(s) = \sum_{i=1}^{IP} |r_1(i) - r_2(i-s)|, \quad (34)$$

where s is the displacement, i the location of an image point along the line considered, IP the number of image points, $r_1(i)$ and $r_2(i)$ the image intensities measured. The value for s for which $F(s)$ is minimal is taken as the image displacement, see fig. 12.

(4) *Reproducibility* is determined by the signal-to-noise ratio in the observed images and can therefore, in principle, be made arbitrarily high by increasing the measuring time.

(5) No *special kind of contrast* in the image is needed to focus the TEM; the BID method functions with both phase and amplitude contrast.

(6) These operations can be carried out with *inexpensive* small computers. For practical reasons it may be wise to use a one-dimensional approach in estimating the image displacement. The amount of data to be processed is small when only one row of pixels in a measured image is used. The

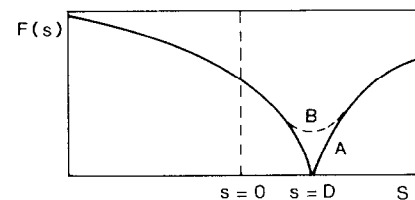


Fig. 12. The function $F(s)$ without noise (A) and with noise (B). D is the displacement between the two images.

Table 2

Summary of experiments; values of parameters are: $\Delta x = 0.5$ nm, $\Delta f = (\Delta f)_{\text{Sch}} = 1$ Sch, $S_{\text{ni}} = 5 \times 10^{-9}$ electrons m, $T = 5$ s, $\Delta \xi = 0.5$ nm, $k = 1250$; dimensions are those at specimen level; 1 Sch = 86 nm

Experiment	Varied
1	Δf : -3.4 to $+3.5$ Sch
2	$\Delta \xi$: 0.1 – 0.2 nm
3	Δx : 0.1 – 7.0 nm
4	$S_{\text{ni}}/(k \Delta x)$: 0.1 – 5.0
5	M : 10^3 – 10^6
6	T : 0.1 – 20.0 s

direction of the image displacement is known, because the shift is in the direction of the beam tilt. Thus when after a certain measuring time the precision is unsatisfactory, more rows of pixels can be processed to obtain a higher precision.

6. Conclusions and discussion

The theoretical limitations of a procedure to autofocus a TEM based on measuring a beam-tilt-induced image displacement (BID) have been studied. Simulations of a TEM and measuring system show that:

- The minimum measuring time necessary to focus the TEM with a precision of 5 nm (under normal operating conditions) with a vidicon camera attached is about 0.05 s. This is the *measuring* time. Image processing time is not taken into account.
- The precision with which the defocus can be estimated depends strongly on the defocus itself. A strategy is proposed for focussing a TEM using this dependence.
- The magnification can be chosen such that the precision in estimating the defocus is optimal. For a TEM under normal operating conditions and normal Vidicon video camera, this magnification is about 200,000.

In studying the theoretical limitations of the BID method, effects of inaccurate modelling of the measuring process are not taken into account. It is assumed, for instance, that the frequency-dependent shifts due to beam tilt in the two images may be neglected.

Furthermore, a well aligned TEM is assumed in studying the theoretical limitations of the BID method. Work is being done in extending the BID method to be able to align the illuminating beam and to apply the BID method to dark field images.

Assumptions have also been made with respect to the power spectrum and modulation depth of the image of the specimen (flat and 0.05, respectively). Automatic focussing with a specimen whose image contains only low frequencies (crystalline) and with a very small modulation depth will be extremely difficult using the BID method.

The time necessary to autofocus a TEM consists of the measuring time necessary to obtain the precision required, and the time necessary to process the measured images and to control the TEM. Limitation of the total autofocussing time is therefore mainly determined by the image pick-up device (afterglow in the fluorescent layer), transport and storage of images, computation time, and settling time of the currents through the beam tilt coils and objective lens currents.

Experiments with the BID method are presently carried out at Delft University of Technology. Results show that the method works under normal operating conditions and that the total time necessary to autofocus the TEM is limited by the processing of the images (10 s). Practical problems arise when calculating the image displacement because of the spatial inhomogeneity of the camera. A system which can align, focus and correct astigmatism using the BID method is being developed.

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Appendix A. Power spectrum of the measured image of the specimen

Suppose $f(x)$, the image on the screen, has a power spectrum $S_{ff}(\omega)$. Our problem is to derive an expression for the power spectrum of $g(x)$, the

image on the screen *measured* with an infinite number of sensors of finite dimensions. In other words, we wish to establish the relation between the power spectra of $f(x)$ and $g(x)$:

$$g(x) \stackrel{\text{def}}{=} \int_{-\Delta x/2}^{\Delta x/2} f(x' - x) dx', \quad (\text{A.1})$$

where Δx is the length of a sensor.

Suppose the expectation of $g(x)$ is $E(g(x)) = \mu_g$ and $E(f(x)) = \mu_f$. Then

$$\mu_g = \int_{-\Delta x/2}^{\Delta x/2} \mu_f dx'. \quad (\text{A.2})$$

$$g(x) - \mu_g = \int_{-\Delta x/2}^{\Delta x/2} [f(x' - x) - \mu_f] dx'. \quad (\text{A.3})$$

With $g'(x) \stackrel{\text{def}}{=} g(x) - \mu_g$ and $f'(x) \stackrel{\text{def}}{=} f(x) - \mu_f$, it follows that

$$g'(x) = \int_{-\Delta x/2}^{\Delta x/2} f'(x' - x) dx'. \quad (\text{A.4})$$

The autocovariance function $C_{gg}(\xi)$ of $g'(x)$ is given by

$$\begin{aligned} C_{gg}(\xi) &= E(g'(x)g'(x + \xi)) \\ &= \int_{-\Delta x/2}^{\Delta x/2} dx'_1 \int_{-\Delta x/2}^{\Delta x/2} dx'_2 E(f'(x'_1 - x) \\ &\quad \times f'(x'_2 - x - \xi)) \\ &= \int_{-\Delta x/2}^{\Delta x/2} dx'_1 \int_{-\Delta x/2}^{\Delta x/2} dx'_2 C_{ff}(x'_2 - x'_1 - \xi). \end{aligned} \quad (\text{A.5})$$

If $x'_2 - x'_1 = r$, then

$$\begin{aligned} C_{gg}(\xi) &= \int_{-\Delta x/2}^{\Delta x/2} dx'_1 \int_0^{\Delta x/2 - x'_1} C_{ff}(r - \xi) dr \\ &\quad + \int_{-\Delta x/2}^{\Delta x/2} dx'_1 \int_{-\Delta x/2 - x'_1}^0 C_{ff}(r - \xi) dr \\ &= \int_0^{\Delta x} C_{ff}(r - \xi) (\Delta x - r) dr \\ &\quad + \int_{-\Delta x}^0 C_{ff}(r - \xi) (\Delta x + r) dr, \end{aligned} \quad (\text{A.6})$$

or

$$C_{gg}(\xi) = \Delta x \int_{-\infty}^{\infty} C_{ff}(p) d_1(\xi - p) dp, \quad (\text{A.7})$$

with

$$d_1(x) = \begin{cases} 1 - \frac{|x|}{\Delta x} & \text{for } |x| \leq \Delta x, \\ 0 & \text{elsewhere.} \end{cases}$$

The power spectrum of $g(x)$ is then given by

$$S_{gg}(\omega) = (\Delta x)^2 S_{ff}(\omega) \text{sinc}^2(\omega \Delta x/2), \quad (\text{A.8})$$

with Δx the sensor length, ω the spatial radian frequency and $S_{ff}(\omega)$ the power spectrum of the image on screen.

Appendix B. Power spectrum of the Poisson noise in the measured image of the specimen

An expression is derived for the power spectrum $S_{pn}(\omega)$ of the Poisson noise for the hypothetical case that a sensor is placed in every point of a selected interval of the x -axis (this is the spatial equivalent of making "continuous-time" observations during a selected time interval).

Suppose sensors of length Δx are present in the interval $0 \leq x \leq L$. This means the noise is covariant over a length Δx . This covariance is first computed. The desired power spectrum is the Fourier transform of this covariance.

Fig. 13 shows two sensors which overlap over a distance $\alpha \Delta x$. Let the expectation of the number of electrons counted by a sensor be k . Then the expectations of these numbers for processes p , q and r are approximately αk , $(1 - \alpha)k$ and αk respectively. The covariance of the electron counts $p + q$ (sensor 1) and $q + r$ (sensor 2) then satisfies

$$E((p + q)(q + r)) = E(q^2) \quad (\text{B.1})$$

since p and r are not covariant. The right-hand member of this equation equals the variance of q and, since q is Poisson-distributed, this variance is



Fig. 13. Two sensors of length Δx overlapping over a distance $\alpha \Delta x$.

equal to $(1 - \alpha)k$. So, since

$$\alpha = |x|/\Delta x, \quad (\text{B.2})$$

the covariance $C_{pn}(x)$ of the electron counts of two sensors at a distance x is described by

$$C_{pn}(x) = \begin{cases} \left(1 - \frac{|x|}{\Delta x}\right)k & \text{for } |x| < \Delta x, \\ 0 & \text{for } |x| \geq \Delta x. \end{cases} \quad (\text{B.3a})$$

$$(\text{B.3b})$$

The power spectrum $S_n(\omega)$ of the Poisson noise is now given by the Fourier transform of $C_{pn}(x)$:

$$S_n(\omega) = k \Delta x \text{sinc}^2(\omega \Delta x/2), \quad (\text{B.4})$$

with ω the spatial radian frequency in the image on screen. Note that in the above considerations the simplification has been made that the standard deviation of the number of electrons counted per sensor is constant; this is true when the modulation depth of the image is small.

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